

Problem Set 2 – PART 1 (part 2 will come next week)

ECON 2010 – Macroeconomics

Spring 2019

The entirety of this problem set is **due at the start of class Week 5**. You have two weeks to complete the problem set.

Note that this problem set will work the same as the first problem set and all problem sets. Work on a separate sheet of paper on your own, and we will correct the problem set together in class on the 14th.

Name:

Problem set instructions.

1. **Write your answers on a separate sheet of paper.** Do not write on this packet until we go over the problem set during class. You will use this problem set sheet to correct the answers that you provide on the separate sheet when we go over the problem set in class the day that it is due.
 2. If possible, please write your final answers in pen.
 3. Please write your name on this packet and on the separate sheet(s) of paper with your answers to the questions.
 4. When appropriate, show your work for each problem. Even if your answer is technically incorrect, you may get many points for the work you did to arrive at that answer. If in doubt, show your work.
 5. Note that not all questions have *one* right answer. Some questions might have many correct answers.
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Problem 1 (59 points)

In this problem, you'll consider the supply of and demand for kale. This probably isn't something you consider often. I certainly don't. But let's see how it goes. If you don't already know, kale is a leafy green that has become quite popular in the last five years. Between 2012 and 2013, sales of kale grew 40% in the US; remarkable for a vegetable. Kale has been called a superfood — because it's high in vitamins and minerals — and a fad food — because at any new or trendy restaurant you go to, you'll inevitably find a kale salad on the menu. Some people think kale is pretty tasty. Some people think everyone who considers kale tasty has lost their mind. But we won't get into that debate.

Instead, consider a market for kale that includes just two consumers — Darius and Georgio — and two producers (growers) — MegaKale Corp. and Kale Yeah!. The demand schedules for each Darius and Georgio are given below, as well as the supply schedules for each of the two producers.

Table 1: Individual demand schedules for kale

Price (per lb.)	Darius' Quantity Demanded (lb.)	Georgio's Quantity Demanded (lb.)
\$4.00	10	0
\$3.00	20	3
\$2.00	30	5
\$1.00	40	7
\$0.00	50	9

Table 2: Individual supply schedules for kale

Price (per lb.)	MegaKale Corp.'s Quantity Supplied (lb.)	Kale Yeah!'s Quantity Supplied (lb.)
\$4.00	62	14
\$3.00	52	10
\$2.00	42	6
\$1.00	32	2
\$0.00	0	0

- (a) Restricting the example to just two consumers and two producers is an example of employing which fundamental principle of economic thinking — that is, which tool from the economists toolbox? **(1 point)** **Remember: Don't write on this problem set. See the problem set instructions.**
- (b) Draw Darius's individual demand curve on an x,y plane. Choose your axes as chosen in class (and as demonstrated in your textbook) such that the demand curve is downward sloping. Be sure to label each of your axes. **(3 points)**

- (c) Draw Georgio's individual demand curve on an x,y plane. Choose your axes as chosen in class (and as demonstrated in your textbook) such that the demand curve is downward sloping. Be sure to label each of your axes. **(3 points)**
- (d) If Darius and Georgio are the only two consumers in the market for kale, draw the market demand curve for kale on a new x,y plane. Choose your axes as chosen in class (and as demonstrated in your textbook) such that the demand curve is downward sloping. Be sure to label each of your axes. **(5 points)**
- (e) Draw the individual supply curve for MegaKale Corp. on an x,y plane. Choose your axes as chosen in class (and as demonstrated in your textbook) such that the supply curve is upward sloping. Be sure to label each of your axes. **(3 points)**
- (f) Draw the individual supply curve for Kale Yeah! on an x,y plane. Choose your axes as chosen in class (and as demonstrated in your textbook) such that the supply curve is upward sloping. Be sure to label each of your axes. **(3 points)**
- (g) If MegaKale Corp. and Kale Yeah! are the only two producers in the market for kale, draw the market supply curve for kale on a new x,y plane. Choose your axes as chosen in class (and as demonstrated in your textbook) such that the supply curve is upward sloping. Be sure to

label each of your axes. (5 points)

- (h) When we model market supply and market demand, we assume that the product in the market we are considering is *undifferentiated*. That is, it is *completely* identical across the market. Producers are producing interchangeable products and consumers demand that undifferentiated/identical product. But what if there was a small difference between the kale grown by each of our two producers? Consider how market supply might possibly change if MegaKale Corp. produces conventional (non-organic) kale and Kale Yeah! produces organic kale. Organic kale costs more to produce because growers have to obtain certifications, pay for inspectors to come to their farms, and use (oftentimes) less effective fertilizers and pesticides. (Hold all else fixed — *ceteris parabis* — as you consider this change.) Would you keep the market supply curve the same as in part (g) above? Or would you draw a different market supply somehow. Do **not** draw or re-draw the market supply. Just state whether it would be the same or different. Then explain why you would expect it to be the same, or change, in a few sentences. (5 points)

- (i) Now consider this one change from part (h) in the context of market demand. If Kale Yeah! changes to producing organic kale, how would this change the market demand curve? Assume that the demand schedules from Georgio and Darius are for non-organic (i.e. conventional) kale. *Important: Hold all else fixed when you consider this change.* Do **not** draw or re-draw the market demand. Just state whether you think that it would be the same or different. Then explain why you would expect it to be the same, or change, in a few sentences. (5 points)

- (j) Now that you've constructed the market demand curve (part (d)) and the market supply curve (part (g)) for kale, draw each together on new x,y plane. Again, remember to label your axes. Use a dotted line from the intersection point of the two curves to indicate the *approximate*

location of the equilibrium quantity. Do the same to indicate the *approximate* location of the equilibrium price. (Note: In order to calculate the equilibrium price and quantity, you would need to use algebra to solve a system of equations. I'm not asking you to do that.) Is the equilibrium price more or less than \$2.25? Is the equilibrium quantity more or less than 49 lb. of kale? (Note: Draw your supply and demand curves carefully to answer these questions.) **(5 points)**

(k) Consider the market equilibrium for kale that you found in part (j) above. Would there be excess supply or excess demand in the market if, *all else equal* (a.k.a. ceteris parabis, holding everything else fixed), Kale Yeah! raised its price per pound by \$0.10? **(3 points)**

(l) Consider the market equilibrium for kale that you found in part (j) above. Would there be excess supply or excess demand in the market if, *all else equal* (a.k.a. ceteris parabis, holding everything else fixed), Darius would only pay \$0.10 *less* per pound than the equilibrium price in the market? **(3 points)**

(m) Consider the market equilibrium for kale that you found in part (j) above. Re-draw that equilibrium (it can be approximate). Then, draw a new market equilibrium on the same plot to reflect a de-regulation of the kale industry. Specifically, consider that before this change, kale producers had to hold up production a few times a month to have government inspectors come make sure their products were safe for consumption. They also had to submit a lot of paperwork that was costly and time-consuming to file. But now, after de-regulation, they no longer are subject to these costly inspections and paperwork. Hold all else fixed besides this increase in the price of fertilizer.

One of your two curves (either market demand or market supply) will stay fixed in your new plot. The other curve will shift. Draw the new shifted curve with a dotted line. (Use solid

lines for the starting equilibrium; i.e. the equilibrium from part (j)). I don't tell you exactly how much the curve should shift, so just draw a reasonable shift given the information above. The important part is the direction of the shift relative to the curve's starting point. Which curve shifts? Is the new equilibrium price for kale higher or lower than it was in part (j)? Is the new equilibrium quantity higher or lower than it was in part (j)? **(7 points)**

- (n) Consider the market equilibrium for kale that you found in part (j) above. Re-draw that equilibrium (it can be approximate). Then, draw a new market equilibrium on the same plot to reflect what happens when the price of spinach decreases. Assume that spinach is a substitute for kale in the market. Hold all else fixed other than the change in the price of spinach when you consider the market for kale.

One of your two curves (either market demand or market supply) will stay fixed in your new plot. The other curve will shift. Draw the new shifted curve with a dotted line. (Use solid lines for the starting equilibrium; i.e. the equilibrium from part (j)). I don't tell you exactly how much the curve should shift, so just draw a reasonable/approximate shift given the information above. The important part is the direction of the shift relative to the curve's starting point. Which curve shifts? Is the new equilibrium price for kale higher or lower than it was in part (j)? Is the new equilibrium quantity higher or lower than it was in part (j)? **(7 points)**

- (o) Consider the following statement: "Because kale is such a healthy vegetable, the government should adopt a price ceiling for kale so that it never becomes too expensive and people will have access to a cheap and healthy vegetable in the grocery store." Is this an example of a positive or a normative statement? **(1 point)**